Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 37)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows some radioactive isotopes, their half-lives and the type of radiation they emit.

Table 1

Isotope	Half-life	Main radiation	Source
Iodine-131	8 days	Beta / gamma	Fission in reactors
Strontium-90	29 years	Beta	Fission in reactors
Caesium-137	30 years	Beta / gamma	Fission in reactors
Plutonium-239	24 400 years	Alpha	Neutron bombardment of U-238
Uranium-238	4.5×10^9 years	Alpha	Occurs naturally in rocks

0 1 . 1 Define the term **half-life**.

[1 mark]

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0 1 . 2 A sample of iodine-131 has an activity of 6400 Bq. Calculate its activity after 32 days.

[2 marks]

Answer ..... Bq

0 1 . 3 Use Table 1 to explain why isotopes with half-lives of several decades, such as caesium-137, cause the greatest concern after a nuclear accident.

[3 marks]

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0 1 . 4 Explain why plutonium-239 is relatively safe outside the body but very dangerous if inhaled or ingested.

[3 marks]

0 1 . 5 Explain what is meant by an **activation product**.

[1 mark]

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0 2 . 1 Explain why it is difficult to be certain that a particular flood or heatwave is caused by anthropogenic climate change.

[3 marks]

0 2 . 2 Explain why historic temperature records collected in towns may overestimate global warming.

[2 marks]

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0 2 . 3 Describe how each of the following sources of proxy data is used to estimate past climates and give one limitation.

[5 marks]

Tree rings (dendrochronology) .....

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Ice cores .....

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0 3 . 1 Explain how each of the following human activities changes the amount of carbon in the atmosphere.

[6 marks]

Ploughing farmland .....

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Draining marshland / peat bogs .....

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Toxic pollution of the sea .....

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0 3 . 2 Give **three** anaerobic environments created by human activities that release methane.

[3 marks]

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0 3 . 3 Explain how rising atmospheric CO_2 causes ocean acidification.

[1 mark]

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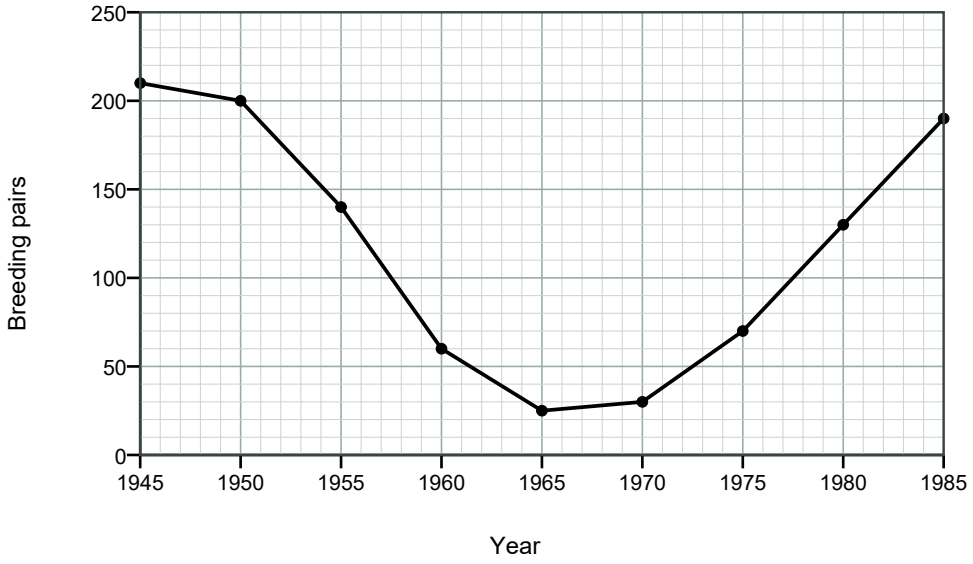
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0 4 Figure 1 shows the number of breeding pairs of sparrowhawks in a study area in the UK between 1945 and 1985, and the years in which DDT use began and was banned.

Figure 1



0 4 . 1 Describe the trend shown in Figure 1.

[2 marks]

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0 4 . 2 Calculate the percentage decrease in breeding pairs between 1950 and 1965. Show your working.

[2 marks]

Answer ..... %

0 4 . 3 DDT use in the UK began in the late 1940s and was restricted from the mid 1960s. Explain how the properties of DDT could account for the pattern in Figure 1.

[4 marks]

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0 4 . 4 Suggest why the recovery after 1970 was slow.

[2 marks]

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0 5 . 1 Which of the following water sources is usually the most convenient for large communities? Tick one box.

[1 mark]

A Desalinated seawater ☐

B River water ☐

C Rainwater catchment ☐

D Newly discovered aquifers ☐

0 5 . 2 Explain why the quality requirements for water differ between uses, using **two** named uses.

[2 marks]

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0 5 . 3 Explain why inter-basin transfers may be used and give **one** disadvantage.

[2 marks]

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0 6 A student investigated the effect of pH on the germination of cress seeds.

0 6 . 1 Explain why cress is a suitable species for this investigation but the results may not apply directly to natural ecosystems.

[2 marks]

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0 6 . 2 Describe how the student should decide the range of pH values to test.

[2 marks]

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0 6 . 3 Give **three** variables that should be standardised and explain how long the experiment should continue.

[4 marks]

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0 6 . 4 The student wants to compare the effect of nutrient concentration on the growth of duckweed. Explain why leaf area is a better measure of growth than fresh mass.

[2 marks]

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0 7 A nuclear fuel reprocessing plant discharges small amounts of radioactive material in liquid effluent to a coastal sea.

0 7 . 1 Explain why it cannot be assumed that the discharge will be diluted to insignificant levels.

[2 marks]

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0 7 . 2 Give **four** factors that would be included in a Critical Pathway Analysis of this discharge.

[4 marks]

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0 7 . 3 Explain how the results of the analysis reduce the cost of environmental monitoring.

[2 marks]

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0 7 . 4 Explain why Critical Group Monitoring measures exposure rather than damage to health.

[2 marks]

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0 8 . 1 Explain **three** advantages of biofuels over other renewable energy resources.

[3 marks]

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0 8 . 2 Explain **three** disadvantages of biofuel crops.

[3 marks]

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0 8 . 3 Describe **two** new biofuel technologies.

[2 marks]

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2 .....

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0 8 . 4 Explain how new low-temperature turbine technology could allow geothermal electricity generation in the UK.

[2 marks]

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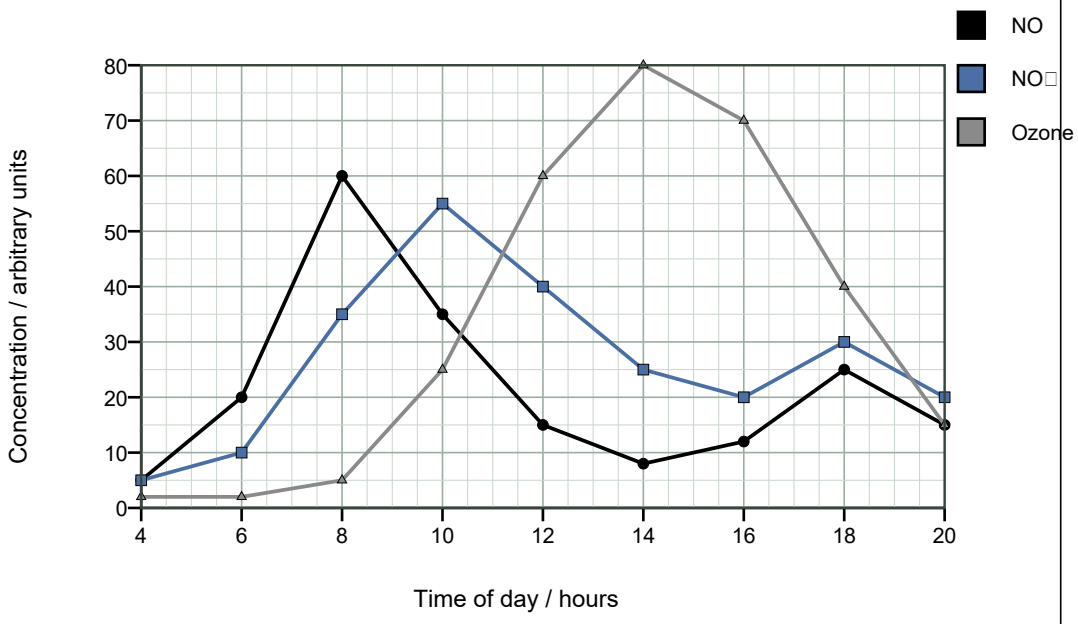
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0 9 Figure 1 shows the concentrations of three pollutants in the air of a large city in a valley during one sunny day.

Figure 1



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0 9 . 1 Use Figure 1 to describe how the ozone concentration changes during the day and explain the pattern.

[4 marks]

0 9 . 2 Explain why the early-morning peak in NO is followed by a peak in NO_2.

[2 marks]

0 9 . 3 Name **one** other toxic secondary pollutant produced in photochemical smogs and give one of its effects on humans.

[2 marks]

0 9 . 4 Explain why cities such as Los Angeles and Mexico City are particularly prone to photochemical smogs.

[2 marks]

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1 0 . 1 Using named examples, explain how the sources of noise pollution can be controlled by changes to design, operation and location.

[9 marks]

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Extra space

1 0 . 2 State what is meant by the **Leq 57 dB** contour used around UK airports.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

[25 marks]

OR

1 1 . 2 Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS